

4. Yeast is used in fermentation to make the ethanol in wine.

(a) Write down the word equation for fermentation. [1]

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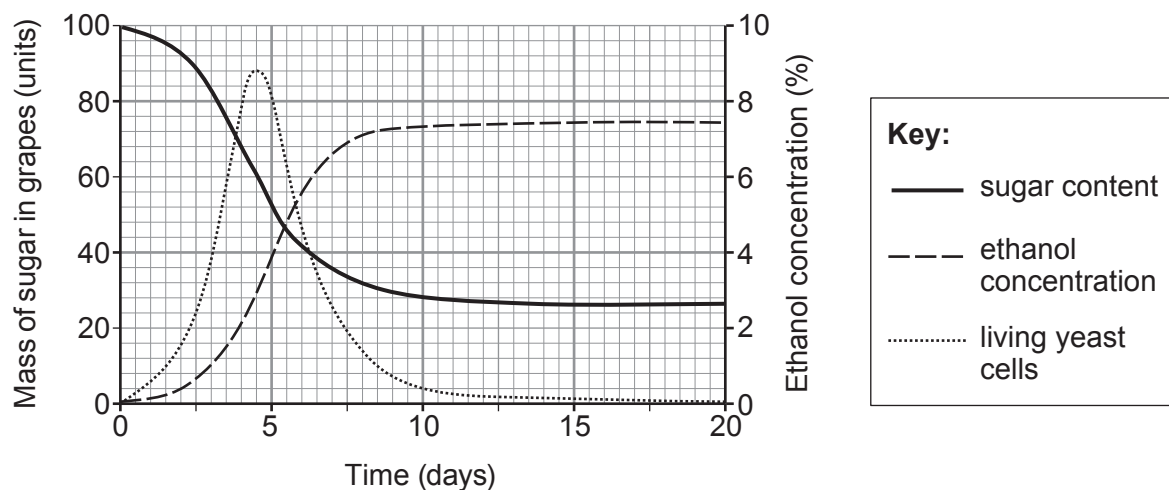
(b) Genetically modified yeast can be used in wine making.

(i) State the meaning of the term 'genetically modified'. [1]

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(ii) One type of genetically modified yeast was used to make wine. The graph below shows how sugar content (solid black line), ethanol concentration (dashed line) and number of living yeast cells (dotted line) change with time.



A home brewer thinks that when the number of living yeast cells is at a maximum, the sugar content will be at its minimum and the ethanol concentration would be at its maximum. Use data from the graph to explain whether you agree. [3]

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- (c) Many countries have banned the use of genetically modified foods.

Suggest **two** reasons for their concern.

[2]

1.

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2.

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- (d) The home brewer also uses this genetically modified yeast to produce beer. Complete the steps in the fermentation process. Three steps have been given. [4]

Step 1: Barley is mixed with water. Starch in the barley is converted to glucose.

Step 2: Hops are added for flavour.

Step 3:

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Step 4:

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Step 5:

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Step 6:

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Step 7: The beer may be pasteurised, then stored in bottles or casks.

